National University of Science and Technology

Atta-ur-Rahman School of Science and Technology

Project Report

INDIGENOUS DEVELOPMENT OF DNA TAGGING SYSTEM FOR INDUSTRIES

May, 2026

Project Approvals

This written technical brief documents the system architecture, operational workflow, mathematical calculations, and empirical qPCR verification of the **DNA Base Tag and Trace System**.

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Chapter 1: Project Background & Objectives

1.1 Introduction

Product counterfeiting and unauthorized tampering pose risks to industrial supply chains, consumer safety, and brand integrity. Conventional security mechanisms, such as surface barcodes and holograms, can be duplicated or physically detached.

To address these challenges, the National University of Sciences and Technology (NUST) developed an **Indigenous DNA Tagging System for Industries**. Using trace quantities of a synthetic DNA construct as a molecular marker, products and packaging can be verified with high specificity using polymerase chain reaction (PCR) and real-time quantitative PCR (qPCR) detection.

1.2 Conventional Security vs. DNA Tagging

Table 1.1 outlines the differences between standard surface markers and synthetic DNA taggants.

Table 1.1. Comparison: Conventional Surface Security vs. Synthetic DNA Tagging

Parameter	Conventional Security (Barcodes, Holograms)	Synthetic DNA Tagging System
Visibility	Overt / Visible surface marker.	**Covert**: Trace molecular marker.
Security Level	Moderate (vulnerable to replication).	**High**: Unique synthetic sequence detected only with specific primers and probes.
Product Integration	Surface label only.	Applied directly onto packaging materials and QR codes.
Verification	Visual inspection or standard optical scan.	**Confirmatory**: PCR amplification and real-time qPCR probe verification.

1.3 Project Objectives

The specific objectives completed in this work include:

1. **Construct Recovery**: Culture and recover the 200 bp synthetic DNA taggant construct supplied in transformed *Escherichia coli* (Stab Top10) in pUC57 plasmid vector.

2. **PCR Amplification and Gel Verification**: Optimize enzymatic amplification conditions using universal primer pairs and verify construct size on 2.0% agarose gel electrophoresis.

3. **Serial Dilution Optimization**: Evaluate serial 1:10 dilutions (D_1 to D_4) of the amplicon to establish optimal working concentrations for qPCR detection.

4. **1.0-Liter Bulk Carrier Formulation**: Prepare a standardized 1.0-Liter formulation of stabilized Tris-EDTA (TE) buffer containing the micro-dosed DNA taggant.

5. **Spatial Homogeneity Assessment**: Verify uniform distribution of the DNA taggant across multiple fractions (F1 to F7) of the 1.0-Liter bulk preparation using real-time qPCR.

6. **Sequence-Specific Probe Verification**: Evaluate four distinct fluorophore probes (Probes A, B, C, and D) in individual and 4-plex multiplex real-time qPCR formats.

7. **Packaging QR Code Application & Recovery**: Apply the formulated taggant onto commercial packaging materials over 2D QR codes and demonstrate recovery via surface swabbing.

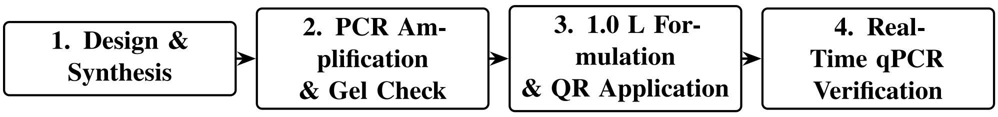


Figure 1.1. Workflow of the Indigenous DNA Tagging System.

Chapter 2: Materials & Taggant Synthesis

2.1 DNA Taggant Construct & Bacterial Host

The molecular marker developed for this project is a unique, synthetic 200 bp double-stranded DNA sequence designed specifically for covert industrial authentication. The construct was supplied cloned into the pUC57 plasmid vector with ampicillin selection (Lot No. BJ0335352-1, miniprep yield ~ 4 μg) and hosted in transformed *Escherichia coli* (Stab Top10).

(a) Cloned Taggant (pUC57) **(b)** Forward Primer Stock **(c)** Reverse Primer Stock

Figure 2.1. Physical synthesis materials: (a) Cloned 200 bp synthetic DNA taggant template in pUC57 plasmid vector (Lot No. BJ0335352-1, ~ 4 μg miniprep), (b) Universal Forward Primer stock (T_m = 58.5°C, reconstitute in 194.4 μL), and (c) Universal Reverse Primer stock (reconstitute in 180.4 μL).

2.2 Bacterial Culture & Colony Heat-Lysis

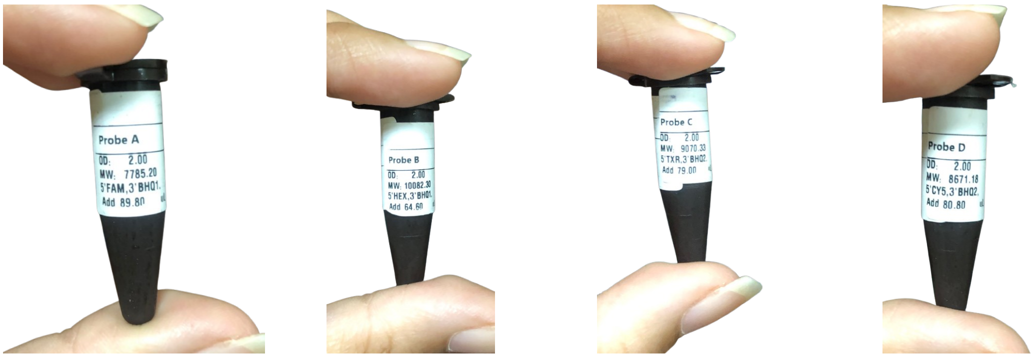
1. **Cultivation**: Transformed Stab Top10 *E. coli* cells were cultured in 10 mL Luria-Bertani (LB) broth supplemented with ampicillin (100 μg/mL) in a shaking incubator at 37°C for 16 hours. Solid cultures were maintained on ampicillin-selective LB agar plates.

2. **Rapid Heat-Lysis**: Rather than utilizing commercial plasmid extraction kits, individual bacterial colonies were suspended in 100 μL of sterile TE buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0). Heat lysis was executed in a thermal cycler at 95°C for 10 minutes. Following a brief centrifugation, the supernatant was used directly as template for PCR amplification.

2.3 Universal Primers & Sequence-Specific Probes

Enzymatic amplification and real-time quantification utilize standardized universal primers and four sequence-specific fluorophore-labeled hydrolysis probes:

- **Universal Forward Primer**: T_m = 58.5°C (reconstituted in 194.4 μL nuclease-free water for 100 μM stock).
- **Universal Reverse Primer**: Reconstituted in 180.4 μL nuclease-free water for 100 μM stock.
- **Sequence-Specific Optical Probes (A, B, C, D)**: Four distinct probes targeting internal regions of the 200 bp taggant construct (Figure 2.2 and Table 2.1).



(a) Probe A (FAM) **(b)** Probe B (HEX) **(c)** Probe C (TxRed) **(d)** Probe D (Cy5)

Figure 2.2. Physical synthesized fluorophore probe stocks: (a) Probe A (5′-FAM, 3′-BHQ1), (b) Probe B (5′-HEX, 3′-BHQ1), (c) Probe C (5′-Texas Red, 3′-BHQ2), and (d) Probe D (5′-Cy5, 3′-BHQ2).

Table 2.1. Physical Specifications of Synthesized 4-Channel Multiplex Probes

Probe	5' Fluorophore	3' Quencher	OD	MW (g/mol)	Reconstitution
Probe A	FAM	BHQ1	2.00	7785.20	Add 89.80 μL
Probe B	HEX	BHQ1	2.00	10082.30	Add 64.60 μL
Probe C	Texas Red (TXR)	BHQ2	2.00	9070.33	Add 79.00 μL
Probe D	Cy5	BHQ2	2.00	8671.18	Add 80.80 μL

2.4 Endpoint PCR Synthesis Formulation & Thermal Cycling

To mass-produce the 200 bp synthetic taggant construct from the bacterial lysate, standard endpoint PCR amplification was conducted in a total volume of 25.0 μL (Figure 2.3, Table 2.2, and Table 2.3).

Thermal Cycling

Step	Temperature	Time	Cycles
Initial Denaturation	95 °C	5 min	1
Denaturation	95 °C	30 sec	
Annealing	55 °C	30 sec	35*
Extension	72 °C	45 sec	
Final Extension	72 °C	7 min	1
Hold	4 °C	∞	1

PCR Recipe

Component	Reaction 1	Reaction 2
10X Buffer	2.5 µL	2.5 µL
$MgCl_2$	1.5 µL	1.5 µL
Forward Primer	1.0 µL	1.0 µL
Reverse Primer	1.0 µL	1.0 µL
Taq DNA Polymerase	1.0 µL	1.0 µL
Template DNA	1.0 µL	2.0 µL
Distilled Water	16.0 µL	15.0 µL
Total Volume	25 µL	25 µL

Figure 2.3. Laboratory master recipe and thermal cycling schedule executed for endpoint PCR amplification.

Table 2.2. Endpoint PCR Synthesis Formulation (25.0 μL Reaction Volume)

Component	Reaction 1	Reaction 2
10× PCR Buffer	2.5 μL	2.5 μL
$MgCl_2$	1.5 μL	1.5 μL
Forward Primer	1.0 μL	1.0 μL
Reverse Primer	1.0 μL	1.0 μL
Taq DNA Polymerase	1.0 μL	1.0 μL
Template DNA	1.0 μL	2.0 μL
Distilled Water	16.0 μL	15.0 μL
Total Volume	**25.0 μL**	**25.0 μL**

Table 2.3. Endpoint Thermal Cycler Parameter Schedule

Step	Temperature	Time	Cycles
Initial Denaturation	95°C	5 min	1
Denaturation	95°C	30 sec	
Annealing	55°C	30 sec	35 cycles*
Extension	72°C	45 sec	
Final Extension	72°C	7 min	1
Hold	4°C	∞	1

*35 cycles applies to the Denaturation, Annealing, and Extension steps (repeated as one cycle).

Chapter 3: Formulation & Packaging QR Code Application

3.1 1.0-Liter Bulk Carrier Formulation in TE Buffer

To enable taggant application, a 1.0-Liter carrier formulation was prepared using sterile Tris-EDTA (TE) buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0). TE buffer maintains DNA stability by chelating divalent metal ions (such as Mg^{2+}).

The master formulation recipe is illustrated in Figure 3.1 and summarized in Table 3.1.

1 Liter of TE Buffer containing taggant is a mixture of

1. 10ml of 1M Tris-HCl
2. 2ml of 0.5M EDTA (pH = 8)
3. 988ml of distilled water
4. 100 ul of PCR product (4-5 tubes of PCR products)

Figure 3.1. Laboratory master batch recipe for the 1.0-Liter stabilized TE buffer taggant carrier formulation.

Table 3.1. 1.0-Liter Bulk TE Buffer Carrier and Taggant Dosing Formulation

Component Reagent	**Dispensed Volume**	**Final Concentration**
1.0 M Tris-HCl (pH = 8.0)	10.0 mL	10.0 mM
0.5 M EDTA (pH = 8.0)	2.0 mL	1.0 mM
Nuclease-Free Distilled Water	988.0 mL	Solvent Matrix
Amplified DNA Taggant (PCR Product)	100.0 μL (4–5 tubes)	Molecular Marker
Total Master Formulation Volume	**1,000.1 mL** ($\approx$ 1.0 **Liter**)	Stabilized Spray Solution

3.1.1 Preparation & Homogenization Procedure

1. **Buffer Preparation**: 10.0 mL of 1 M Tris-HCl and 2.0 mL of 0.5 M EDTA were added to 900 mL of distilled water, adjusted to pH = 8.0, and topped up to 1000 mL.

2. **Autoclaving**: The buffer was sterilized in an autoclave at 121°C for 20 minutes and cooled to room temperature.

3. **Taggant Dosing**: 100.0 μL of amplified 200 bp DNA taggant product (pooled from 4–5 tubes of PCR reactions) was introduced into the 1.0-Liter container.

4. **Homogenization**: The container was mixed on a laboratory blood mixer to achieve uniform spatial distribution throughout the 1.0-Liter volume.

3.2 Application & Recovery on Packaging QR Codes

The formulated 1.0-Liter DNA taggant solution was applied and recovered from commercial packaging materials:

- **Application**: The taggant was applied over printed 2D QR codes and barcodes on commercial pharmaceutical packaging surfaces (cartons, bottles, and blister foils).

- **Recovery Protocol**: A sterile cotton swab moistened with sterile TE buffer was firmly swabbed across the marked QR code surface, and eluted in a microcentrifuge tube for direct qPCR verification.

Chapter 4: Experimental Results & Verification

4.1 Agarose Gel Electrophoresis Verification

To verify the molecular size and fidelity of the synthesized taggant construct, amplified products from gradient PCR (54°C–59°C) were evaluated on a 2.0% agarose gel stained with ethidium bromide (Figure 4.1).

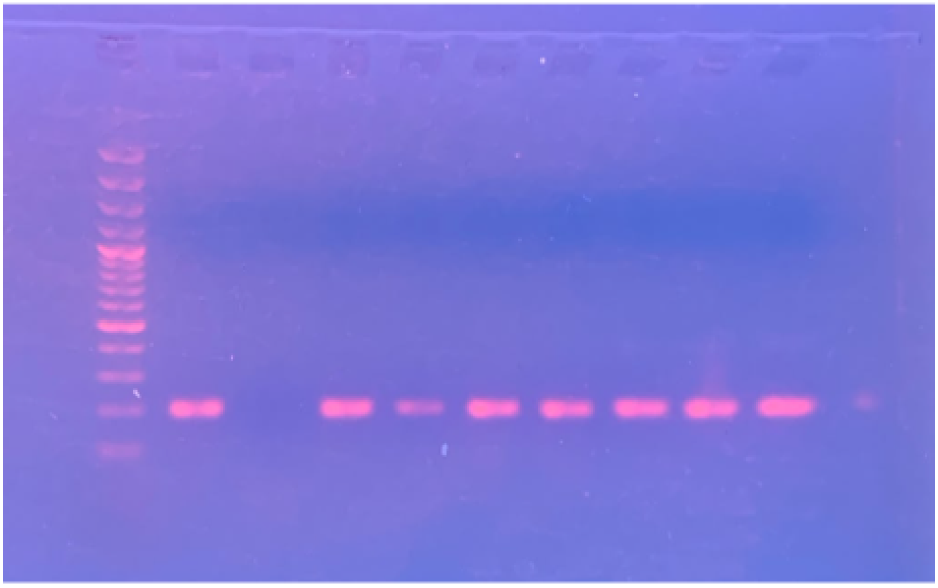


Figure 4.1. Empirical 2.0% agarose gel electrophoresis photograph under UV transillumination showing the DNA ladder on the left (Lane M) and the discrete, high-yield 200 bp amplified DNA taggant bands across gradient annealing lanes.

The electrophoretic run confirmed:

- A single, sharp amplicon band migrated precisely at the 200 bp reference ladder rung.
- Optimal amplification yield was achieved at annealing temperatures between 55°C and 57°C.
- Total absence of non-specific secondary amplicons, primer-dimer complexes, or chimeric artifacts, verifying the high specificity of the universal primer pair.

4.2 Serial Dilution Analysis & Template Dynamics

Real-time qPCR exhibits high analytical sensitivity. In early laboratory testing, undiluted, concentrated PCR product showed partial reaction inhibition. To resolve this and establish the optimal working concentration, serial 1:10 dilutions (D_1 to D_4) were evaluated:

- The concentrated template caused partial kinetic suppression during real-time amplification.
- Serial dilution systematically relieved inhibition, with the 1:10,000 dilution (D_4) producing sharp, reproducible quantification cycles ($C_q \approx 4.78$).
- Consequently, the 1:10,000 dilution was established as the standardized working concentration for bulk carrier formulation.

4.3 1.0-Liter Bulk Homogeneity & Product Recovery

To assess whether the DNA taggant remains uniformly dispersed throughout the 1.0-Liter TE buffer carrier without settling or precipitation, multiple independent fractions (F1 to F7) and product recovery swabbing samples (PR1–PR2) were evaluated via real-time qPCR (Table 4.1).

Table 4.1. qPCR Assessment of Spatial Homogeneity Across 1.0-Liter Bulk Reagent

Sample Fraction	Template Volume	Observed C_q	Homogeneity Status
F1	1.0 μL	**4.21**	Positive (Uniform Dispersion)
F2	1.0 μL	**5.03**	Positive (Uniform Dispersion)
F3	1.0 μL	**5.75**	Positive (Uniform Dispersion)
F4	1.0 μL	**5.34**	Positive (Uniform Dispersion)
F5 (Replicates R1/R2)	1.0–2.0 μL	**5.03 / 6.01**	Positive (Uniform Dispersion)
F6 (Replicates R1/R2)	1.0–2.0 μL	**5.75 / 6.59**	Positive (Uniform Dispersion)
F7 (Replicates R1/R2)	1.0–2.0 μL	**5.34 / 5.36**	Positive (Uniform Dispersion)
PR (Replicates R1/R2)	1.0–2.0 μL	**3.99 / 5.23**	Positive (Product Recovery)

The consistent cycle threshold values (C_q = 3.99–6.59) across all sampled fractions and swabbed packaging surfaces confirm uniform liquid dispersion and stable recovery of the DNA taggant.

4.4 Real-Time qPCR Formulation & Cycling Parameters

Forensic detection was standardized to a 20.0 μL reaction volume utilizing commercial real-time master mix, universal primers, and sequence-specific probes (Figure 4.2,

Table 4.2, and Table 4.3).

PCR Recipe

Component	Reaction 1	Reaction 2
Master Mix	10.0 µL	10.0 µL
Forward Primer	0.6 µL	0.6 µL
Reverse Primer	0.6 µL	0.6 µL
Template DNA	1.0 µL	2.0 µL
ddH_2O	7.8 µL	6.8 µL
Total Volume	20 µL	20 µL

Thermal Cycling

Step	Temperature	Time	Cycles
Initial Denaturation	95 °C	5 min	1
Denaturation	95 °C	30 sec	
Annealing	55 °C	30 sec	35*
Extension	72 °C	45 sec	
Final Extension	72 °C	7 min	1
Hold	4 °C	∞	1

**35 cycles applies to the Denaturation, Annealing, and Extension steps (repeated as one cycle).*

Figure 4.2. Laboratory master recipe and thermal cycler parameter schedule for real-time qPCR verification.

Table 4.2. Real-Time qPCR Reaction Formulation (20.0 μL Total Volume)

Component	Reaction 1	Reaction 2
Master Mix (2×)	10.0 μL	10.0 μL
Forward Primer	0.6 μL	0.6 μL
Reverse Primer	0.6 μL	0.6 μL
Template DNA	1.0 μL	2.0 μL
Distilled Water (ddH_2O)	7.8 μL	6.8 μL
Total Volume	**20.0 μL**	**20.0 μL**

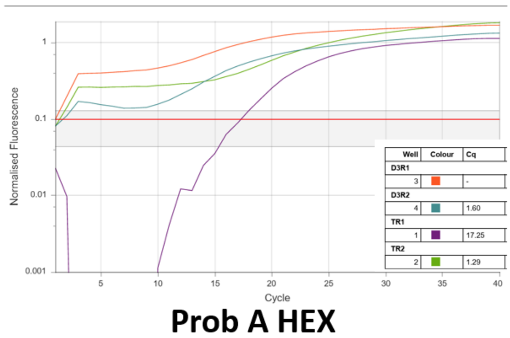
Table 4.3. Real-Time Thermal Cycler Parameter Schedule

Step	Temperature	Time	Cycles
Initial Denaturation	95°C	5 min	1
Denaturation	95°C	30 sec	
Annealing	55°C	30 sec	35 cycles*
Extension	72°C	45 sec	
Final Extension	72°C	7 min	1
Hold	4°C	∞	1

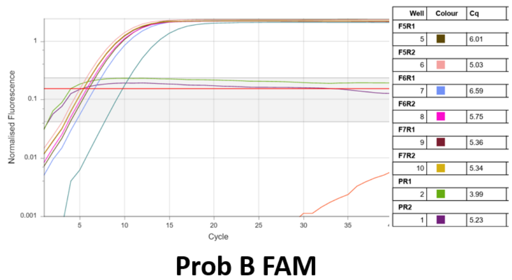
*35 cycles applies to the Denaturation, Annealing, and Extension steps (repeated as one cycle).

4.5 Real-Time qPCR Probe Verification Results

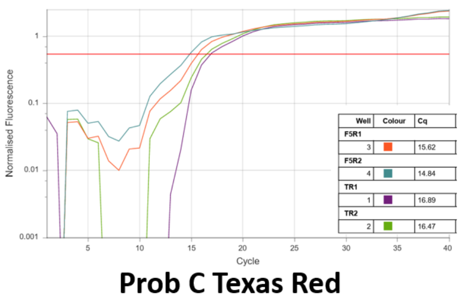
Real-time qPCR amplification was evaluated across all four individual fluorophore probe channels, followed by a simultaneous 4-plex multiplex reaction. Figure 4.3 displays the amplification curves and quantification cycle (C_q) data generated directly by the real-time thermal cycler.



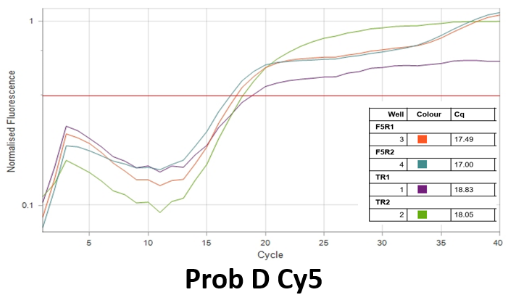
(a) Probe A (HEX Channel)



(b) Probe B (FAM Channel)



(c) Probe C (Texas Red Channel)



(d) Probe D (Cy5 Channel)

Figure 4.3. Empirical real-time qPCR amplification curves and quantification cycle (C_q) data tables captured directly from the thermal cycler across all four optical channels: Probe A (HEX), Probe B (FAM), Probe C (Texas Red), and Probe D (Cy5).

Table 4.4 summarizes the quantification cycle ranges and analytical outcomes across all individual channels and the multiplex format.

Table 4.4. Summary of Empirical Real-Time qPCR Optical Results

Assay Format	Optical Channel	Ex / Em (λ)	Observed C_q Range	Status
Probe A	HEX	535/556 nm	**1.29 – 17.25**	Positive
Probe B	FAM	495/520 nm	**3.99 – 6.59**	Positive
Probe C	Texas Red	585/610 nm	**14.84 – 16.89**	Positive
Probe D	Cy5	650/670 nm	**17.00 – 18.83**	Positive
4-Plex Multiplex	4 Channels	Multi-Channel	**11.95 – 15.49**	Positive

The positive amplification across all four probe channels in both individual and multiplex configurations confirms that the synthetic DNA taggant can be authenticated with complete molecular specificity.

Chapter 5: DNA Encapsulation

Chapter 6: Software Ecosystem

6.1 Overview

The DNA Tag and Trace software ecosystem links the bioinformatic design, automated manufacture, tracking, and laboratory verification through a central database (Figure 6.1).

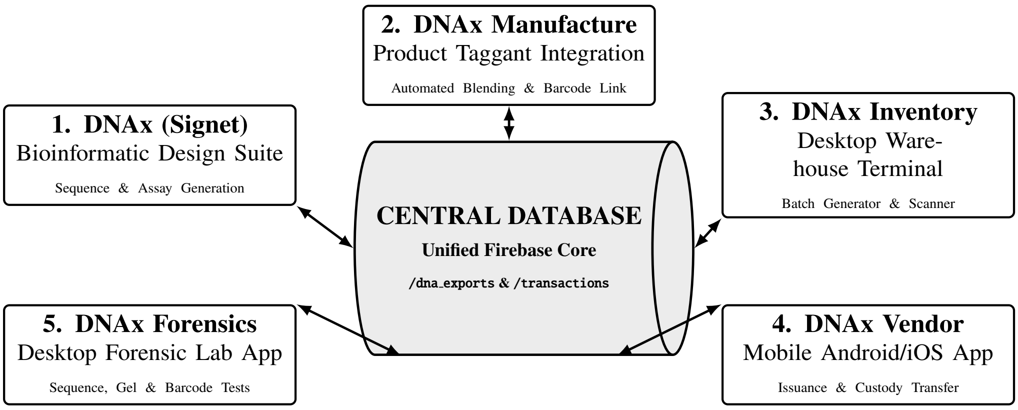


Figure 6.1. The software modules connected through the central database.

6.2 Software Modules

6.2.1 1. DNAx Signet (Bioinformatic Design Suite)

Used for designing synthetic DNA sequences and assay parameters:

- Generates sequence constructs with specified lengths and GC content.
- Calculates melting temperatures (T_m) for primers and probes.
- Screens sequences against standard databases to ensure specificity.
- Exports design specifications and associated QR codes.

6.2.2 2. DNAx Manufacture (Product Integration & Taggant Blending)

DNAx Manufacture is designed to integrate the DNA taggant directly with the product:

- **Product Integration**: Integrates the taggant with the physical product.
- **Automated Combinations**: Automates the combination of taggants from all available taggants while maintaining anonymity across the entire tagging process.

- **Data Association**: Saves all data regarding the specific combination of taggants against each unique barcode.
- **Development Status**: This feature is currently in the development phase, with active R&D underway to complete it end-to-end.

6.2.3 3. DNAx Inventory (Inventory Management)

Desktop module for managing inventory:

- Generates and assigns identifiers to product batches.
- Integrates barcode scanning for logging stock intake and dispatch.
- Tracks inventory records in the database.

6.2.4 4. DNAx Vendor (Distribution & Verification)

Mobile application for distribution checks:

- Scans barcodes/QR codes to verify recorded batches.
- Logs product issuance and handover events.

6.2.5 5. DNAx Forensics (Laboratory Verification)

Specialized verification software for testing laboratories:

- Matches sequence data and gel electrophoresis sizing (e.g., 200 bp).
- Records and verifies qPCR results against registered batch profiles.

6.3 Conclusion

This project successfully established the laboratory workflow for the indigenous DNA tagging system:

- Design of the 200 bp synthetic DNA construct and primers.
- Enzymatic amplification via PCR and size confirmation on 2.0% agarose gel.
- Formulation of the 1.0-Liter TE buffer taggant solution.
- Specific detection and verification across individual and multiplex real-time qPCR probe assays.
- Application and non-destructive swabbing recovery on packaging QR codes.